

GAURAV KAMBOJ

Backend Engineering | Forward Deployed Engineering | Agentic AI

📞 +91-9646914195 | ✉️ gauravkamboj.dev@gmail.com | 🌐 gkamboj | 📄 kambojgaurav

7+ years building backend systems and forward-deployed agentic AI solutions, currently shipping enterprise AI products at SAP serving 100+ customers. Designed and shipped production agentic workflows, semantic search systems, and catalog automation handling 1M+ products. Previously delivered e-commerce systems for global enterprises with measurable business outcomes.

SKILLS

Languages: Go, Python, Java
Generative AI: LangGraph, LangChain, FastMCP, RAG, Agentic AI, Vector Embeddings, Prompt Engineering
Tools & Frameworks: gRPC/Protobuf, GraphQL, Spring Boot, Kafka
Datastores: PostgreSQL, SAP HANA, Redis
Cloud & Infrastructure: SAP BTP, Azure, Jenkins, Kubernetes, Docker

EXPERIENCE

SAP Labs

Gurgaon, IN

SDE-2

September 2021 - Present

CX AI Toolkit: Enterprise AI platform serving 100+ customers

- Designed and developed the **Catalog Optimization Agent** using LangGraph, automating product attribute population and description generation for commerce catalogs with up to **1M+ products**, replacing a fully manual enrichment process.
- Built the **AI Tools Builder**: Go gRPC service serving custom tool definitions to agents at runtime, with GraphQL APIs for tool management, Trie-based schema traversal for entity extraction, and prompt validation against customer data schemas.
- Built the **Semantic Search** microservice: Go GraphQL gateway proxying to a Python LangGraph ReAct agent over gRPC, combining vector embeddings (HNSW ANN search on SAP HANA) with TOON encoding for natural-language-to-SQL generation over structured business data.
- Built the **Communication Insights** microservice: LangGraph agent crawling Outlook/Gmail mailboxes, applying PII obfuscation, and generating insights from email threads. Integrated Azure Event Hub for real-time email change notifications with cursor-based pagination.
- Owned LangGraph and LangChain major version upgrades across multiple services, managing breaking-change migrations with backward compatibility. Managed OSS compliance (BlackDuck, Checkmarx) across releases.

Forward Deployed Engineering: Agentic AI solutions for global enterprises

- Designed collections workflows for two global FMCG enterprises using LangGraph state graphs with HITL approval gates, email drafting, and SAP FSCM integration for contact recording. Built custom checkpoint for pause/resume across approval cycles.
- Led architecture and design** for a Fortune 500 chemical company's procurement automation: agent-driven order confirmation extraction from 1000+ daily supplier emails, PO discrepancy detection, and automated reconciliation.

E-commerce platform implementations

- Integrated payment gateways (PayU, Ozow, PayGate) for South Africa's leading pharmacy chain, handling credit cards, EFT, and digital wallets with retry logic and refund flows.
- Led a team of four to implement location-based pricing for India's largest B2B fashion platform (3M+ products, 1L+ sellers), building multi-hierarchy pricing and multi-warehouse fulfillment engine, **increasing seller revenue by 15%** and **reducing shipping costs by 50%**.
- Built backend services for Asia's largest theme park, enabling ticket booking and dynamic pricing across campaigns.

Nagarro Software

Gurgaon, IN

SDE-1

January 2019 - September 2021

- Developed core microservice components for a high-volume car finance platform using Spring Boot, processing 1000+ daily loan applications with multi-lender and credit check integrations.
- Designed and optimized REST APIs through query restructuring and caching, achieving 35% reduction in P95 response time. Implemented JUnit and Mockito test coverage with Splunk-based monitoring for production observability.

EDUCATION

Thapar Institute of Engineering & Technology, Patiala, IN

July 2014 - July 2018

Bachelor of Engineering in Computer Engineering

Lovely Professional University, Phagwara, IN

July 2021 - February 2023

Master of Science in Information Technology (Distance)